

Machine Learning Day 2

Types of Machine Learning Systems

In broad, there are 5 main types of machine learning systems that are:

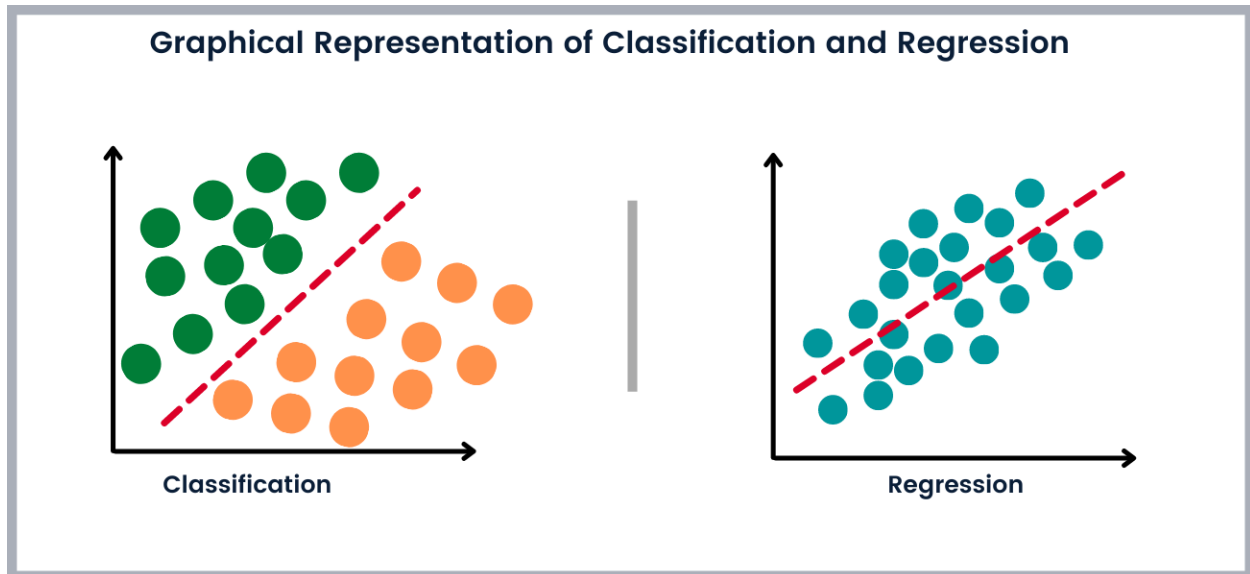
- Supervised learning
- Unsupervised learning
- Semi-supervised learning
- Self-supervised learning
- Reinforcement learning

Supervised Learning

Most machine learning tasks fall into supervised learning type. As the name implies, a supervised learning model is trained with input data along with some form of guidance that we can call labels. In other words, a supervised learning model maps the input data (or X in many textbooks) to output labels (y). Labels are also known as targets and they act as a description of the input data.

The example of facemask recognition that we used in above sections is a good example of supervised learning. In broad, there are 2 main kinds of supervised learning problems that are:

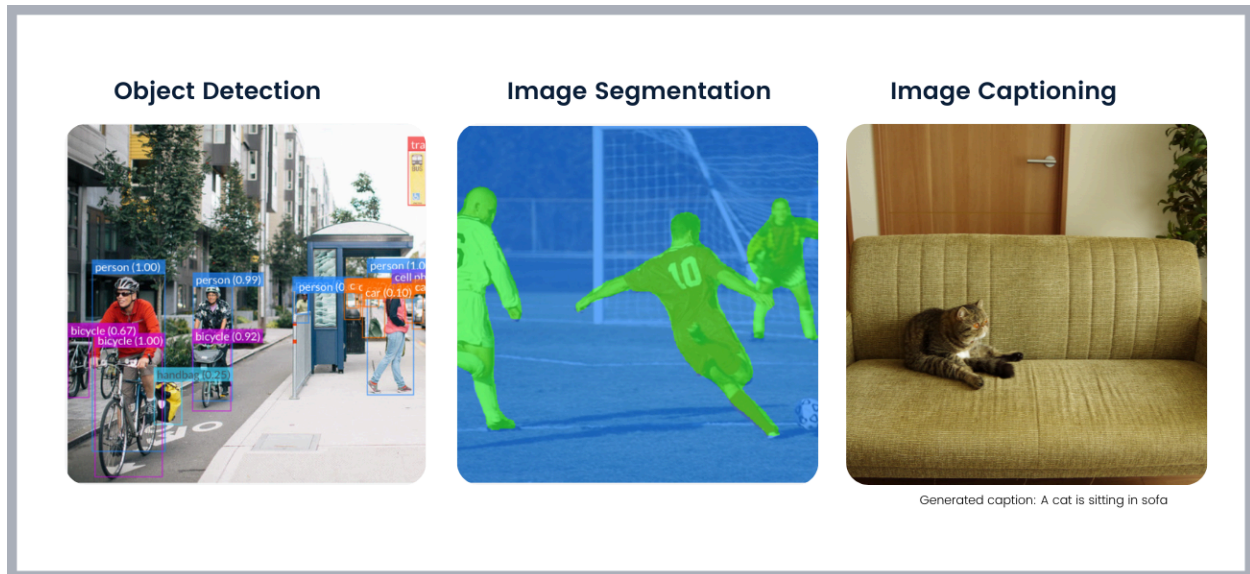
- **Classification problems** where the task is to identify a given category from numerous categories or simply make choice between a number of categories. Another example of classification task is to identify if the incoming email is either spam or not based on the email contents.
- **Regression problems** where the goal is to predict a continuous value of something. A classical example for this category is to predict the price of the used car given its features such as brand, age, number of doors, number of sits, safety level, maintenance cost, etc...



Supervised learning algorithms includes shallow algorithms such as **linear and logistic regression, decision trees, random forests, K-Nearest Neighbors(KNN),** and **support vector machines(SVM)**. Neural networks can be both supervised(like using them for image classification) and unsupervised.

With that said, there are other advanced tasks that falls into supervised learning type such as:

- Image captioning where the goal is to predict the caption of a given image.
- Object detection where the goal is to recognize the object in image and draw the bounding box around it.
- Image segmentation where the goal is to group the pixels that make up particular objects in the image.



Some of those tasks can involve both classification and regression. Take an example for object detection, a task of recognizing and localizing an object in an image: it involves classification (recognizing the object among many other objects) and regression (predicting the coordinates of the objects in an image to make a bounding box).

If any of the things we are talking about sounds unfamiliar, do not worry. There is no way to explain all things at once, but as we progress, things will get clear.

Unsupervised Learning

Supervised learning algorithms are trained with data and labels. Conversely, unsupervised learning algorithms are trained on **unlabelled data**.

Unsupervised learning algorithms are primarily used for:

▼ Clustering: K-Means

Clustering is the process of dividing a dataset into groups (called clusters) such that:

- Data points in the **same cluster are similar**
- Data points in **different clusters are dissimilar**
- **Dimension reduction and data visualization:** Principal Component Analysis (PCA), t-Distributed Stochastic Neighbor Embedding (t-SNE).

There are many more algorithms that could also be used for those 2 tasks, but, we will cover **K-Means**, **PCA**, and **t-SNE** because they are commonly used.

Semi-supervised learning

Semi-supervised learning falls between supervised and unsupervised learning. In semi-supervised learning, a small portion of training data is labeled while the rest of the data points are not labeled.

Labeling data is one of the most challenging things in machine learning. Because it requires a skilled human, it's expensive and there is always a chance of getting the labels wrong. Thus, having a way to make use of unlabelled data can improve the results of a particular task with a minimal need of labels, and ultimately reducing the time and cost that would be needed to label all the training data.

Semi-supervised learning is most notable in problems that involve working with massive datasets like internet image searches, image and audio recognition, and webpages classification. As you would imagine for example, no body that is responsible for a boring task of labelling millions of images that are uploaded daily on social media platforms like Instagram.

Another example: According to [sitefy](#), 252.000 new websites are created everyday. As you can also imagine, it would not be a good use of time and resources to assign a task of labelling these websites to someone when semi-supervised learning can remove such overheads.

Self-supervised learning

Self-supervised learning is one of the most exciting types of machine learning that is most applicable in computer vision and robotics. While semi-supervised learning uses a small portion of labeled data, self-supervised learning uses entire unlabeled data and it does not require manual annotations, removing the need for humans in the process.

Producing a dataset with good labels is expensive, while unlabeled data is being generated all the time. The motivation of self-supervised learning is to make use of the large amount of unlabeled data. The main idea of self-supervised learning is to generate the labels from unlabeled data, according to the structure or characteristics of the data itself, and then train on this unsupervised data in a supervised manner.

Reinforcement Learning

Reinforcement learning is a special type of machine learning that is most applicable in robotics and games.

In reinforcement learning, a learning system called *an agent* can perceive the environment, performs some actions, and gets *rewarded* or *penalized* depending on how it is performing. The main goal of the agent is to accumulate as much as rewards as possible.

The *agent* learns the best strategy(*policy*) necessary for getting the most reward itself.

Let's summarize the types of machine learning systems. By far, supervised and unsupervised learning are the two commonly used types of machine learning. Semi-supervised and self-supervised learning are also getting attractions in deep learning community, but they are still in research. We will focus more those two practical types: supervised and unsupervised.

The Typical Machine Learning Workflow

Although every machine learning problem is unique, they all follow a similar workflow. In this section, we will learn how to approach machine learning problems systematically.

Overall, a typical machine learning project workflow consists of:

- Defining and formulating a problem
- Collecting data
- Establishing a baseline
- Exploratory data analysis(EDA)
- Preparing data
- Selecting and training a model
- Performing error analysis and improving a model
- Evaluating a model
- Deploying a model

Defining a Problem

Everything starts here. Problem definition is the important and the initial step in any machine learning project. This is where you make sure you understand the problem really well. Understanding the problem will give you proper intuitions about the next steps to follow such as right learning algorithms, etc. But wait, what does it mean to understand the problem?

Understanding the problem is all about diving deep into the details of the problem at hand and asking the right questions. First, it's important to narrow down the problem until you can have a simple and a well defined goal. Here are examples of simple goals: To *classify* products into different categories, to *predict* the price of a used car given its features (such as brand, age, etc...), to *recognize* if a person is wearing a facemask, to *divide* customers into different groups that share similar behaviors, etc...As you can see, the goal can tell whether the problem is classification, regression, or clustering...

At this stage, avoid vague words. The simpler you can formulate the problem, the better things will be down the hall. It's also worth questioning if the project should be pursued. Most problems do not require machine learning.

Problem definition also goes beyond determining the goal and objective of the project to thinking about the data that will be needed. Machine learning models relies on the data, and the better models comes from better data. Do you have data that contains the things you want to predict? Models are not magical things, they are mathematical functions that takes data along with labels, and determine the patterns that can be used to make predictions on unseen data. If the data does not contain meaningful information relevant to what you want to predict, you will get poor predictions.

After you understand the problem and you have an idea of the data you want, the next step is to collect it.